PROJECT REPORT

Retail Business Performance and Profitability Analysis

Sales, gross profit and inventory insights using SQL, Python and Power BI

Prepared by	Muhammed Bishar Ashraf N
Report date	October 2026
Data period	1 January 2022 – 6 February 2022 (37 days)
Sources	SQL script (retail_analysis.sql), Python notebook (Retail_Inventory_Analysis.ipynb), Power BI dashboard (.pbix and PDF export)
Tools	SQL · Python (pandas, matplotlib, seaborn) · Power BI Desktop

Please read before using the numbers

All results cover **1 January – 6 February 2022** only. February is a **partial month** (six days), so no January-versus-February comparison in this report should be read as a full-month or seasonal trend.

Inventory-day figures are **estimates built on just 37 days of sales**.

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Executive Summary

This report looks at revenue, gross profit and inventory for a retail chain of 50 stores in Mexico that sells 35 products in five categories. It brings together three pieces of work: SQL views, a Python correlation analysis and a Power BI dashboard. **All results cover 1 January – 6 February 2022 (37 days).**

$651.88K	$199,311	30.57%	−0.178
Revenue (Power BI card)	Gross profit (sales analysis)	Overall gross margin	Correlation (849 rows)

Key findings

- **Toys earned the most revenue** ($237,440.69), but **Electronics earned the most gross profit** ($80,509) and had the **highest gross margin** (48.18%). The overall sales gross margin was about 30.57%.
- **Downtown stores produced roughly 58% of revenue** (about 0.38M of 0.65M on the dashboard). Airport stores had the lowest total but, with only 3 stores, the highest revenue per store.
- **Inventory-day estimates exist for 849 of 1,593 store-product rows.** The other 744 are NULL, meaning no sales were recorded in the period. NULL does not mean zero, and these rows still hold stock that is not moving.
- **The Python correlation between estimated inventory days and gross profit is −0.178** across 849 rows: a weak negative relationship that does not prove inventory causes profit changes.
- **Two gross-profit totals appear** ($199,311 in the sales analysis and $198,825 in the inventory analysis). They come from different tables and should not be treated as identical.

Three cautions that apply to every finding

- **February is incomplete.** Only 1–6 February is included, so the January-to-February drop on the dashboard is not a full-month or seasonal comparison.
- **Inventory days rest on 37 days of sales.** They are rough estimates of stock cover, not forecasts.
- **Some totals differ between files.** A recalculation from the uploaded CSV files comes out about 2% higher than the dashboard (Appendix A explains what could and could not be confirmed).

1. Project Objective and Business Questions

Objective

The aim of this project is to understand **how the business is performing and where profit and inventory risk sit**. It does this by measuring sales, gross profit and stock levels, and then presenting the results in a dashboard that a manager can read quickly.

The work is deliberately beginner-friendly: each measure is defined in plain language (see Appendix B), and every result is tied back to the file it came from.

Business questions

1. **Which product categories** bring in the most revenue, gross profit and gross margin?
2. **How did revenue change** between January and the first days of February in the data?
3. **Which store locations** (Downtown, Commercial, Residential, Airport) generate the most revenue?
4. **How many days of sales** does the stock on hand cover for each store and product?
5. **Is there a relationship** between estimated inventory days and gross profit?
6. **Which products and stores** need inventory attention first?

> **Scope of the analysis**
>
> Period analysed: **1 January – 6 February 2022 (37 days).** The Power BI date slicer is set to 01-01-2022 to 06-02-2022 (day-month-year).
>
> Because February holds only six days, this is a short-term snapshot, not a full-year or seasonal study.

2. Dataset and Tools Used

2.1 The data

The project uses four data tables, plus a data dictionary that describes their fields. Cost and price are in US dollars.

Table	What it contains	Rows in file	Key fields
Products	Product list with category, cost and retail price	35	Product_ID, Product_Name, Product_Category, Product_Cost, Product_Price
Stores	Store list with city (in Mexico) and location type	50	Store_ID, Store_Name, Store_City, Store_Location, Store_Open_Date
Sales	One row per product sold in a store on a date	829,262*	Sale_ID, Date, Store_ID, Product_ID, Units
Inventory	Stock on hand for each store-product pair	1,593	Store_ID, Product_ID, Stock_On_Hand

The uploaded sales.csv holds transactions from 1 January 2022 to 30 September 2023. The analysis in this report uses only the first 37 days (1 January – 6 February 2022). The data dictionary also lists a Calendar table, but no separate calendar file was uploaded

The tables connect through shared IDs: Sales links to Products through **Product_ID** and to Stores through **Store_ID**, and Inventory links to both. Store locations fall into four types: Downtown (29 stores), Commercial (12), Residential (6) and Airport (3).

2.2 Measures used in this report

Measure	How it is calculated	Plain-language meaning
Revenue	Units sold × product price	Money coming in from sales
Gross profit	Revenue − (units sold × product cost)	Money left after paying for the goods
Gross margin	Gross profit ÷ revenue	Share of each sales dollar kept as profit
Avg units per day	Units sold ÷ 37 days	Typical daily sales pace
Estimated inventory days	Stock on hand ÷ avg units per day	How many days the stock would last at that pace

These formulas match the sample rows shown in the Python notebook (checked by hand in Section 3.2).

2.3 Tools

Tool	Files	Role in the project
SQL	retail_analysis.sql	Holds the query that reads the inventory profit view (inventory_profit_analysis). The model behind the dashboard also uses a sales_analysis table.
Python	Retail_Inventory_Analysis.ipynb	Jupyter notebook using pandas, matplotlib and seaborn to calculate and plot the correlation.
Power BI Desktop	Retail_Business_Performance_Dashboard.pbix and .pdf	The dashboard deliverable: KPI cards, charts, filters and an inventory table.

3. Data Cleaning and Validation in SQL

3.1 What the SQL script contains

The uploaded SQL file, **retail_analysis.sql**, contains a single statement, “SELECT * FROM inventory_profit_analysis;”, and it appears twice on one line. This statement simply displays the finished inventory view.

> **Evidence gap**
> The statements that **create** the views (inventory_profit_analysis and sales_analysis), including any cleaning or filtering rules, are **not included** in the uploaded files. This report therefore does not claim specific SQL cleaning steps. What it can show is what the view outputs reveal (Section 3.2) and what independent checks on the raw CSV files found (Section 3.3).

3.2 What the SQL view output shows

The notebook loads the view output (1,593 rows and 8 columns). The columns and their behaviour are:

Column	What the output shows
Store_ID, Product_ID	One row per store-product pair in the inventory data (1,593 rows, the same count as inventory.csv).
Product_Name, Product_Category	Descriptive fields taken from the Products table.
Stock_On_Hand	Units in stock (a single snapshot, not a history).
Avg_Units_Per_Day	Units sold divided by 37 days.
Estimated_Inventory_Days	Stock on hand divided by average units per day. Shown as NULL when there were no sales.
Gross_Profit	Profit on the units sold in the period (shown as 0 when nothing sold).

> **Worked example (Store 1, Chutes & Ladders)**
> Stock on hand is 6 units. Five units were sold over 37 days, giving 5 ÷ 37 ≈ **0.14 units per day**. Stock cover is 6 ÷ 0.135 ≈ **44.4 days**, which is exactly what the view shows. Gross profit is 5 units × ($12.99 price − $9.99 cost) = **$15**, which also matches.

3.3 Validation checks on the source files

The checks below were run **for this report** (in Python, on the uploaded CSV files). They are not taken from the SQL script, but they show whether the data is safe to analyse.

Check	Result	Why it matters
Missing values in all four tables	**None found**	Gaps would distort totals
Duplicate transactions (Sale_ID)	**None found**	Duplicates would double-count sales
Sales rows pointing to a missing product or store	**None found**	Every sale can be linked to a price and a location
Duplicate or negative inventory rows	**None found** (lowest stock is 0)	Stock levels are usable
Products priced above cost	**All 35 products**	No product sells at a loss
Price and cost stored as text (e.g. “$9.99 ”)	**Conversion needed**	Must be turned into numbers before calculating revenue
Store-product pairs that sold but have no inventory row	**2 pairs** (Store 5: Classic Dominoes and Mini Basketball Hoop; 57 units)	Can be left out of an inventory-based view (see Section 4.5)
Date coverage of sales.csv	**1 Jan 2022 – 30 Sep 2023**	The analysis uses only the first 37 days

3.4 Handling rules the outputs are consistent with

- **Convert** price and cost from text to numbers.
- **Filter** sales to 1 January – 6 February 2022 (37 days).
- **Join** Sales to Products (for price, cost and category) and to Stores (for location).
- **Keep NULL as NULL** for stock-cover estimates when there are no sales, instead of replacing it with 0 (Section 7.1).

4. Revenue, Gross Profit and Gross Margin by Product Category

4.1 Headline results

These are the reported figures from the sales analysis and the Power BI dashboard.

Result	Value	Source
Total revenue	$651.88K	Dashboard card
Total gross profit	about $199,311	Sales analysis (dashboard card shows 199K)
Overall gross margin	about 30.57%	Sales analysis
Highest revenue: Toys	$237,440.69	Sales analysis
Highest gross profit: Electronics	$80,509	Sales analysis
Highest gross margin: Electronics	48.18%	Sales analysis

4.2 Category rankings on the dashboard

The three category charts on the dashboard agree on the order of the categories. Unit totals are labelled on the chart; revenue and gross profit are shown as bars without labels, so only their ranking is read from the dashboard.

Rank	By revenue	By gross profit	By units sold
1	Toys	Electronics	Toys (13.1K)
2	Electronics	Toys	Electronics (10.4K)
3	Games	Games	Games (10.3K)
4	Sports & Outdoors	Sports & Outdoors	Sports & Outdoors (6.5K)
5	Art & Crafts	Art & Crafts	Art & Crafts (5.3K)

4.3 Full category breakdown

Because the dashboard does not label every bar, the table and chart below were **recalculated from the uploaded CSV files** for 1 January – 6 February 2022. They come out about 2% higher than the reported figures (see Appendix A) but show the same ranking and almost the same margins.

Category	Units	Revenue	Gross profit	Margin	Share of revenue	Share of profit
Toys	13,401	$242,791.99	$51,334	21.14%	36.5%	25.2%
Electronics	10,653	$170,721.47	$82,215	48.16%	25.7%	40.4%
Games	10,459	$115,854.41	$33,671	29.06%	17.4%	16.6%
Sports & Outdoors	6,660	$92,787.40	$23,963	25.83%	13.9%	11.8%
Art & Crafts	5,465	$43,225.35	$12,240	28.32%	6.5%	6.0%
Total	**46,638**	**$665,380.62**	**$203,423**	**30.57%**	**100%**	**100%**

Recalculated from products.csv, stores.csv and sales.csv for 1 Jan – 6 Feb 2022. Reported figures are in Section 4.1.

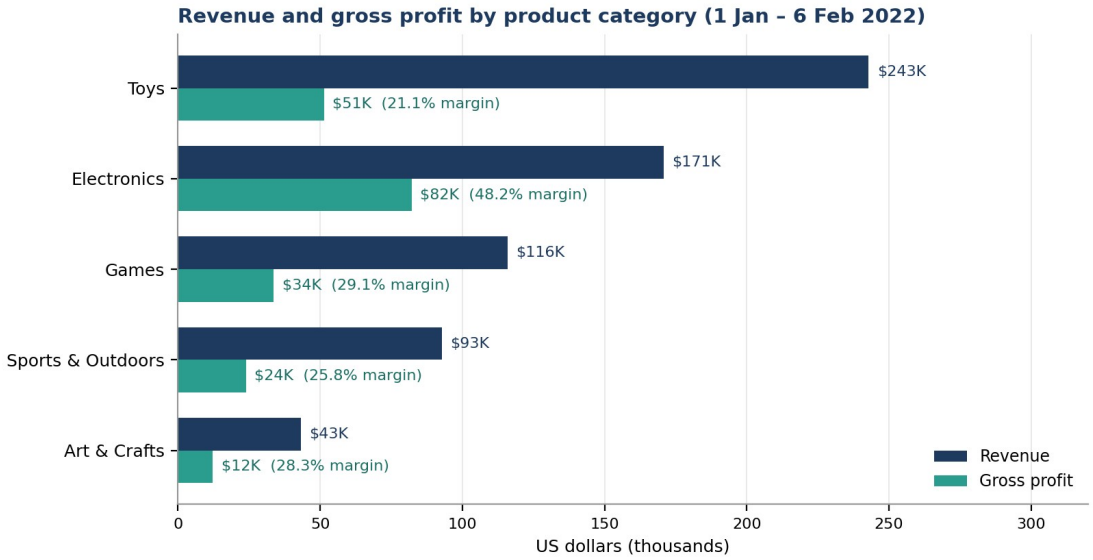


Figure 1. Revenue and gross profit by product category (recalculated from the CSV files).

4.4 How to read the results

- **Revenue and profit tell different stories.** Toys sells the most but keeps about 21 cents of profit per sales dollar. Electronics sells less, yet keeps about 48 cents, so it produces roughly 40% of all gross profit from about 26% of revenue.
- **Games, Sports & Outdoors and Art & Crafts** sit in the middle on margin (about 26–29%) and are smaller contributors to total profit.
- **Margins agree closely across sources.** The overall margin is 30.57% in both the reported and recalculated figures, and Electronics is 48.18% reported versus 48.16% recalculated.

4.5 Why there are two gross profit totals

The dashboard shows two gross-profit totals that are close but **not the same**:

Total	Amount	Where it comes from
Sales analysis	about $199,311	Built from sales transactions (sales_analysis table); feeds the gross profit card
Inventory analysis	about $198,825	Built from the inventory view (one row per store-product pair); feeds the inventory table total and the gross-profit-by-category chart
Difference	**about $486** (0.24%)	Small, but real

Possible explanation (not confirmed)

An inventory-based view only includes store-product pairs that have an inventory record. In the CSV files, two pairs sold in the period but have no inventory record (Section 3.3). Recalculated from the CSVs, their gross profit is about $534, which is similar in size to the $486 gap. The view definitions were not provided, so this could not be verified.

What to do with this: use the sales-analysis total for company-wide gross profit, and use the inventory-analysis total only when working with inventory data. The two should not be treated as identical.

5. Monthly Results and Store-Location Findings

5.1 The Power BI dashboard

The dashboard is the visual deliverable of the project. It combines KPI cards, a date slicer, store-location and product-category filters, five charts and an inventory table.

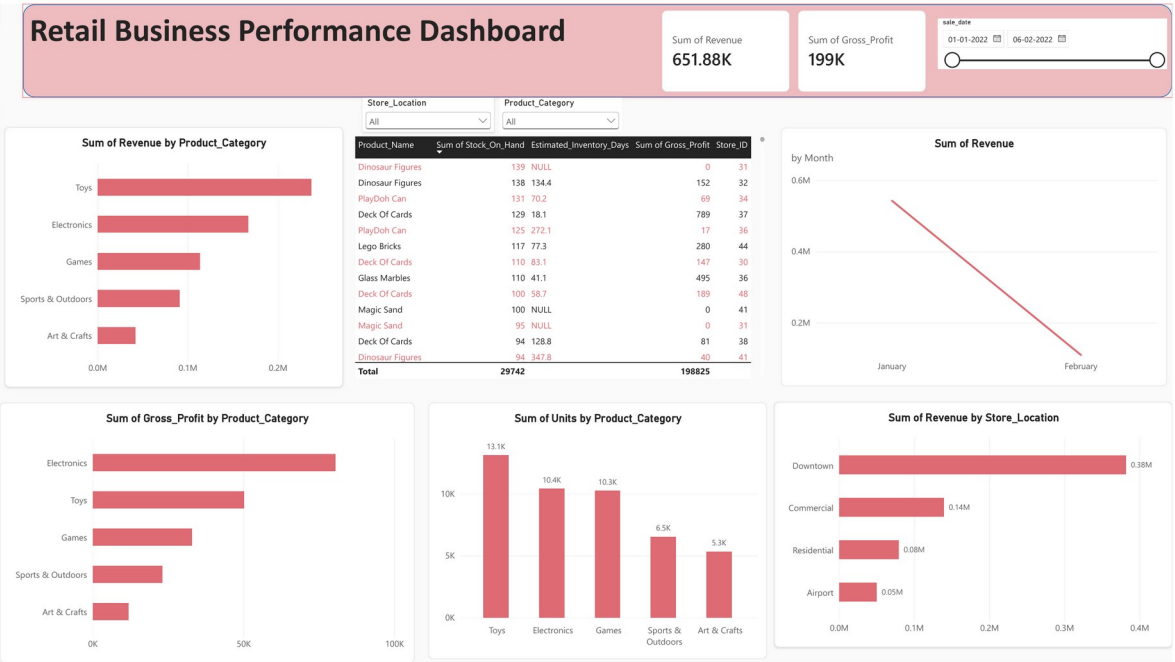


Figure 2. Retail Business Performance Dashboard (Power BI export, filters set to All, dates 01-01-2022 to 06-02-2022). Zoom in for detail; the full-size export is Retail_Business_Performance_Dashboard.pdf.

Each visual reads from one of two tables in the Power BI model:

Dashboard visual	Source table
Revenue and gross profit cards	sales_analysis
Revenue by product category; units by product category	sales_analysis
Revenue by month; revenue by store location	sales_analysis
Gross profit by product category	inventory_profit_analysis
Inventory table (stock, estimated days, gross profit, store)	inventory_profit_analysis

5.2 Monthly results

The dashboard's monthly chart shows revenue starting at roughly **0.54M in January** and falling to roughly **0.1M in February**. The chart has no data labels, so these are approximate readings.

> **Do not read this as a real decline or a seasonal pattern**
> January has **31 days** of data but February has only **6 days (1–6 February)**. A line that drops from a full month to six days will always fall steeply, whatever sales were actually doing. This is a **partial-month artefact**, not a full-month or seasonal comparison.

A fairer rough check divides revenue by the number of days. The figures below were recalculated from the CSV files:

Period	Days of data	Revenue	Revenue per day	Share of window
January 2022	31	$542,554.91	$17,502	81.5%
February 2022 (1–6 only)	6	$122,825.71	$20,471	18.5%

Recalculated from the CSV files, so the totals are about 2% above the dashboard (Appendix A).

Per day, the first six days of February ran slightly above the January average, but six days is far too short to call this growth, and daily revenue swings widely (recalculated: from about $11.6K on the quietest day to about $30.8K on 5 February). **The honest conclusion is that this data cannot show a January-to-February trend.** A complete February is needed first.

5.3 Store-location findings

Store location	Number of stores	Revenue (dashboard)	Approx. share of revenue	Approx. revenue per store
Downtown	29	0.38M	58%	$13K
Commercial	12	0.14M	22%	$12K
Residential	6	0.08M	12%	$13K
Airport	3	0.05M	8%	$17K
Total	**50**	**0.65M**	**100%**	**$13K**

Revenue from the dashboard's store-location chart (rounded to 0.01M). Store counts from stores.csv. Shares and per-store values are approximate because the dashboard values are rounded.

- **Downtown leads in total revenue** mainly because it has the most stores (29 of 50), not because each store sells much more.
- **Airport stores look the most productive per store** (about $17K each versus $12–13K elsewhere), but there are only three of them, so this is a small sample.
- **Commercial stores earn slightly less per store** than Downtown and Residential stores.
- The dashboard shows revenue by location only. Gross profit by location is not shown, so no profit conclusion about locations is drawn here.

6. Python Analysis: Estimated Inventory Days and Gross Profit

6.1 What the notebook does

1. Loads the exported inventory view (**1,593 rows, 8 columns**) using pandas.
2. Keeps two columns, Estimated_Inventory_Days and Gross_Profit, and **drops rows with a missing (NULL) estimate**.
3. Calculates the correlation between the two columns, using pandas' default method.
4. Draws a scatter plot with seaborn and matplotlib.

Item	Value
Rows in the inventory view	1,593
Rows with an estimate (used in the correlation)	**849**
Rows left out because the estimate is NULL	744 (1,593 − 849)
Correlation: estimated inventory days vs gross profit	**−0.178**

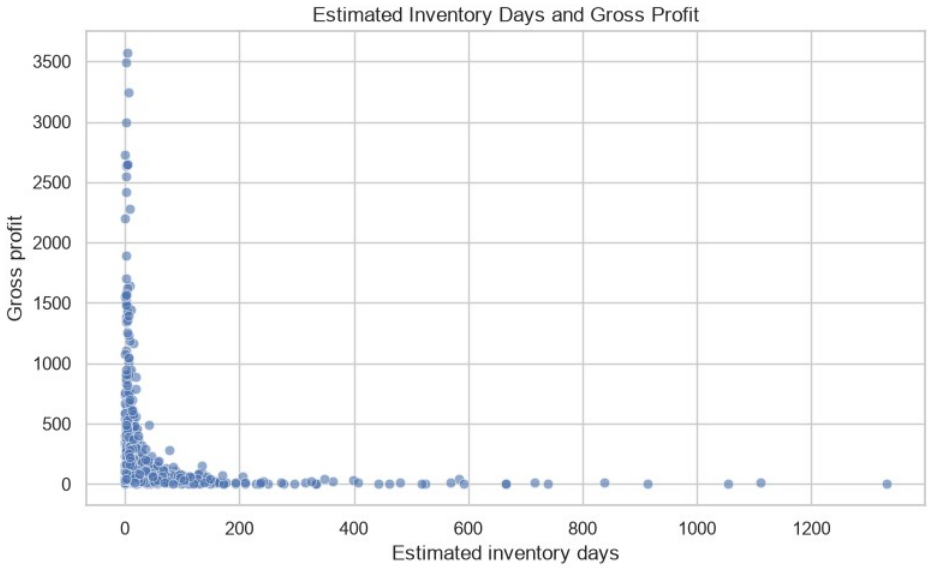


Figure 3. Estimated inventory days and gross profit, 849 store-product rows (output of the Python notebook).

6.2 How to read −0.178

A correlation runs from −1 to +1. A value near 0 means two measures barely move together; values near −1 or +1 mean they move together strongly. At **−0.178**, the relationship is **weak and negative**: rows with more days of stock tend to have slightly lower gross profit, but the pattern is loose.

The scatter plot shows why. The highest-profit rows sit at low inventory days (fast sellers), and the very high day-estimates (several hundred days and more) all have gross profit close to zero. But most rows are crowded together near the bottom left, so the overall link is weak.

> **This is not proof that inventory causes profit changes**
>
> Estimated inventory days are **calculated from sales**, so the two measures are not independent: a product that sells quickly naturally shows fewer days of stock and also earns more profit. The correlation describes how the numbers move together, not what causes what.
>
> It also only covers the 849 rows that have an estimate. The 744 NULL rows, which have no sales, are not part of it (Section 7).

6.3 The 37-day limitation

Every estimate here assumes that the sales pace seen over **37 days** continues. That is a short window that includes a partial February, and it takes no account of promotions, seasons or deliveries. Treat the estimates as a rough guide for deciding what to look at, not as precise forecasts.

7. Inventory Findings and What NULL Means

7.1 What does a NULL estimate mean?

> **Plain-language definition**
> Estimated inventory days = stock on hand ÷ average units sold per day. If **no units were sold** in the 37 days, the daily sales pace is zero and the division cannot be done. The result is **NULL**: *not calculable*.
>
> **NULL means no sales were recorded for that store-product combination during the period.** It is not zero days, and the rows should not be thrown away without explanation.

The three situations below look similar in a spreadsheet but mean very different things (examples are real rows from the notebook, all in Store 1):

Product (Store 1)	Stock	Sold in 37 days	Estimated days	What it means
Chutes & Ladders	6	Yes (≈0.14/day)	44.4	Normal: stock lasts about 44 days at the current pace
Classic Dominoes	0	Yes (≈0.51/day)	0.0	Sold out but still selling: possible lost sales
Barrel O' Slime	32	No sales	NULL	Stock is sitting unsold: cannot be calculated, and not the same as 0

Why NULL matters. A value of 0.0 days warns that a product is running out. NULL warns about the opposite problem: stock that is not moving at all. Treating NULL as 0 would hide exactly these rows, and dropping them hides the products most likely to be over-stocked. They were left out of the Python correlation only because a correlation cannot use missing values, which also means that correlation says nothing about them.

7.2 What the dashboard's inventory table shows

The table lists store-product rows from the largest stock down. These are the first 13 rows visible in the dashboard export:

Product	Stock on hand	Estimated days	Gross profit	Store ID
Dinosaur Figures	139	NULL	$0	31
Dinosaur Figures	138	134.4	$152	32
PlayDoh Can	131	70.2	$69	34
Deck Of Cards	129	18.1	$789	37
PlayDoh Can	125	272.1	$17	36
Lego Bricks	117	77.3	$280	44
Deck Of Cards	110	83.1	$147	30
Glass Marbles	110	41.1	$495	36

Product	Stock on hand	Estimated days	Gross profit	Store ID
Deck Of Cards	100	58.7	$189	48
Magic Sand	100	NULL	$0	41
Magic Sand	95	NULL	$0	31
Deck Of Cards	94	128.8	$81	38
Dinosaur Figures	94	347.8	$40	41

Dashboard totals: 29,742 units in stock; table gross profit total $198,825.

- **Three of the 13 rows are NULL** (Dinosaur Figures in Store 31, Magic Sand in Stores 41 and 31). Together they hold 334 units that recorded no sales.
- **Very long cover appears elsewhere:** PlayDoh Can in Store 36 (272.1 days, $17 profit) and Dinosaur Figures in Store 41 (347.8 days, $40 profit).
- **High stock is not automatically a problem.** Deck Of Cards in Store 37 holds 129 units, yet covers only 18.1 days and earned $789, because it sells.

7.3 The wider inventory picture

To size the issue beyond the visible rows, the figures below were **recalculated from the CSV files** using the same 37-day method. Total stock (29,742 units) and the count of 849 rows with an estimate match the dashboard and notebook; the remaining counts may differ slightly from the SQL view (Appendix A).

Measure	Result
Store-product rows in inventory	1,593
Rows with an estimate / NULL rows	849 / 744
NULL rows that still hold stock	690 rows, about 10,400 units (about 35% of all stock)
NULL rows with no stock and no sales	54 rows
Rows with zero stock (of which sold something in the period)	77 (23)
Rows with more than 90 days of cover / more than 180 days	93 / 36
Middle (median) estimate across the 849 rows	about 17 days

NULL rows are more common in some categories than others:

Category	Rows in inventory	NULL rows	Share NULL
Art & Crafts	397	251	63%
Sports & Outdoors	323	150	46%
Toys	447	195	44%
Electronics	149	59	40%
Games	277	89	32%

Recalculated from the CSV files. A high NULL share can mean many products are slow sellers in many stores, or that stores carry products they do not actually sell; the data cannot tell which.

8. Practical Recommendations

The recommendations below follow directly from the findings. Items 1–3 are inventory actions that can start now. Items 4–6 are about commercial focus. Items 7–9 improve the quality of future analysis.

#	Recommendation	Evidence
1	**Review NULL items first.** Check whether they are discontinued, misplaced, wrongly recorded or simply unpopular in that store. Then consider moving stock to stores where the product sells, discounting, or pausing reorders.	About 744 NULL rows, holding roughly a third of all stock (Section 7.3). Examples: Dinosaur Figures (Store 31), Magic Sand (Stores 41 and 31).
2	**Review products with very long stock cover** before reordering. Set a target range for days of cover and flag rows that go far beyond it.	PlayDoh Can, Store 36: 272.1 days; Dinosaur Figures, Store 41: 347.8 days (Section 7.2).
3	**Check for stock-outs on items that are selling.** Replenish those first to avoid lost sales.	Rows with 0.0 days, e.g. Classic Dominoes in Store 1. About 23 such rows were found (Section 7.3).
4	**Protect Electronics.** Keep it well stocked and visible, since it delivers the most gross profit and the best margin.	$80,509 gross profit and 48.18% margin, the highest of any category (Section 4.1).
5	**Look closely at Toys margin.** Review supplier cost, pricing and discounting, because small margin gains on the biggest-revenue category matter most.	Toys has the highest revenue ($237,440.69) but ranks second for gross profit (Section 4.2); its margin is about 21% in the recalculation.
6	**Learn from Airport stores, and keep Downtown well supplied.** Study what drives Airport sales per store; keep availability high where most revenue is earned.	Downtown is about 58% of revenue; Airport is about $17K per store versus $12–13K elsewhere (Section 5.3).
7	**Re-run the analysis on a longer period** (several months, with complete months compared like for like) before making large ordering decisions.	Current estimates rest on 37 days and a partial February. The uploaded sales.csv runs to September 2023, so longer windows are possible.
8	**Improve the dashboard.** Mark February as partial, add a gross-margin measure, add a note explaining NULL, and label which gross-profit total each visual uses.	Monthly chart is easy to misread (Section 5.2); two gross-profit totals appear (Section 4.5).
9	**Save the full SQL in retail_analysis.sql**, with comments, including the statements that create both views and any cleaning rules.	The uploaded script holds only a SELECT statement (Section 3.1), so the cleaning logic cannot be reviewed or reproduced.

9. Limitations and Conclusion

9.1 Limitations

- **Short window.** The data covers 1 January – 6 February 2022 (37 days). Longer-term and seasonal patterns cannot be assessed.
- **Partial February.** Only six days of February are included, so month-to-month changes are not full-month comparisons.

- **Estimates built on 37 days.** Estimated inventory days assume the recent sales pace continues, and ignore promotions, deliveries and seasonality.
- **One stock snapshot.** Stock on hand is a single point in time, not a history, so it is unclear when stock arrived or whether items were out of stock earlier.
- **Weak, non-causal correlation.** The −0.178 correlation covers 849 rows, leaves out the 744 NULL rows, and cannot show cause and effect.
- **Two gross-profit totals.** $199,311 (sales analysis) and $198,825 (inventory analysis) differ and should not be treated as identical. The cause could not be confirmed.
- **About 2% gap to the raw CSV files.** A recalculation from the uploaded CSVs is higher than the dashboard, and a few individual rows differ (Appendix A). Reported figures are used as the main results.
- **SQL logic not visible.** The uploaded SQL script does not show the view definitions or cleaning steps.
- **Gross profit only.** Gross profit ignores rent, wages, shrinkage and other operating costs, so it is not net profit.
- **Dashboard readings.** Some dashboard values (monthly revenue, store-location revenue) are read from unlabelled or rounded charts and are approximate.

9.2 Conclusion

What this analysis tells us

In the first 37 days of 2022, the business earned about **$651.88K in revenue** and about **$199.3K in gross profit** (a margin of roughly **30.57%**). **Toys leads on revenue, but Electronics leads on gross profit and margin**, so the best commercial focus depends on whether the goal is sales volume or profit. Downtown stores generate the most revenue overall, while Airport stores earn the most per store.

On inventory, the clearest risk is **stock that is not selling**: 744 of 1,593 store-product rows have no sales in the period (NULL), and they hold a large share of total stock. The weak correlation of −0.178 across 849 rows suggests that days of stock cover and profit are only loosely related, and it does not show that inventory causes profit changes.

These findings are an **early snapshot**, not a final verdict. February is only partly covered, inventory days rest on 37 days of sales, and a few totals differ between sources. The most valuable next step is to repeat the analysis over a longer, complete period and to save the full SQL so every number can be reproduced.

Appendix A. Data Reconciliation Notes

The brief for this report asked that differences between files be pointed out and explained rather than guessed. This appendix records them.

A.1 Reported figures versus a recalculation from the CSV files

Reported figures come from the Power BI dashboard, the SQL views and the notebook. The recalculation applies the formulas in Section 2.2 to the uploaded CSV files for 1 January – 6 February 2022.

Measure	Reported	Recalculated	Difference	Difference %
Total revenue	$651.88K	$665,380.62	+$13.5K	+2.1%
Total gross profit (sales)	$199,311	$203,423	+$4,112	+2.1%
Overall gross margin	30.57%	30.57%	0.00 pts	–
Toys revenue	$237,440.69	$242,791.99	+$5,351	+2.3%
Electronics gross profit	$80,509	$82,215	+$1,706	+2.1%
Electronics gross margin	48.18%	48.16%	−0.02 pts	–
Inventory-view gross profit	$198,825	$202,889	+$4,064	+2.0%

A.2 What matches and what does not

- **Matches:** total stock of 29,742 units (dashboard) equals the CSV inventory total; the 849 rows with an estimate (notebook) equals the recalculated count; the category rankings are the same; 12 of the 13 inventory rows visible on the dashboard and 4 of the 5 notebook sample rows match the CSV exactly.
- **Does not match:** Lego Bricks in Store 44 shows 77.3 days and $280 gross profit on the dashboard but 59.3 days and $365 from the CSV. Animal Figures in Store 1 shows about 0.27 units per day and $30 in the notebook, but 12 units and $36 from the CSV.
- **Margins agree to within 0.02 percentage points,** so prices and costs appear consistent. The gap is in the **volume of sales captured** (about 2% fewer units behind the reported figures).

A.3 What this suggests, and what is unknown

The pattern suggests the data behind the SQL views, notebook and dashboard holds slightly fewer sales rows than the uploaded sales.csv. The files provided do not show why: the SQL script contains no filtering logic, and the Power BI file stores its data in a compressed format that was not examined at row level. This report therefore uses the reported figures as the main results, uses recalculated figures only where the reported ones are not available (and labels them), and recommends re-running the SQL against the uploaded CSV files to settle the difference.

Appendix B. Glossary for Beginners

Term	Meaning
Revenue	Total money received from sales: units sold × price.
Gross profit	Revenue minus the cost of the goods sold. It does not include rent, wages or other running costs.
Gross margin	Gross profit as a percentage of revenue. A 30% margin means 30 cents of every sales dollar is left after the cost of goods.
Stock on hand	Units currently held in a store.
Estimated inventory days	How many days current stock would last if sales continued at the recent daily pace.
NULL	A database value meaning “no value / cannot be calculated”. It is not the same as zero.
Correlation	A number from −1 to +1 showing how closely two measures move together. Near 0 means weak. It never proves cause.
Store-product row	One product in one store (for example, Dinosaur Figures in Store 31).
View	A saved SQL query that behaves like a table (for example, inventory_profit_analysis).
Partial month	A month with only some of its days in the data. Its totals cannot be compared with a full month.
KPI	Key performance indicator: a headline number such as total revenue.

Appendix C. Source Files and How They Were Used

File	How it was used in this report
retail_analysis.sql	Reviewed. It contains only a SELECT on inventory_profit_analysis (Section 3.1).
Retail_Inventory_Analysis.ipynb	Source of the 1,593-row view structure, the 849-row correlation of −0.178, the scatter plot and sample rows.
Retail_Business_Performance_Dashboard.pdf / .pbix	Source of the headline dashboard figures, charts and inventory table. The .pbix model also identified the two source tables and which visual uses each.
sales.csv, products.csv, stores.csv, inventory.csv	Used to validate the data (Section 3.3) and to recalculate figures the dashboard does not label. Recalculated figures are always marked.
data_dictionary.csv	Used for field definitions, the US-dollar currency and the Mexican store cities.